Predictive Modeling and Risk Scoring for Bank Customer Churn

1. Introduction

Customer churn refers to the situation where customers stop using a bank's services. In the banking sector, retaining customers is very important because acquiring new customers is more expensive than retaining existing ones. This project focuses on identifying customers who are likely to churn using data analysis and visualization techniques.

2. Objective

The main objective of this project is:
•To analyze customer behavior
• To identify customers who are likely to churn
• To segment customers based on their activity and value
• To assign a risk score to each customer

3. Dataset Description

The dataset used in this project contains customer-related information such as:
•Customer ID
• Age
• Account Balance
• Tenure
• Number of Transactions
• Activity Level
• Churn Status
• Risk Score
• Customer Segment

4. Methodology

4.1 Data Collection
The dataset was collected in Excel format and imported into Power BI for analysis.

4.2 Data Preprocessing
•Cleaned the dataset
• Checked for missing values
• Structured the data for analysis

4.3 Customer Segmentation
Customers were divided into different groups such as:
•High Value Customers
• Inactive Customers
• New Customers
• Frequent Users

4.4 Risk Scoring
Each customer was assigned a risk score based on their likelihood to churn:
0–30 → Low Risk
30–70 → Medium Risk
70–100 → High Risk

5. Dashboard and Visualization

A Power BI dashboard was created to visualize the data. The dashboard includes:
•Pie Chart showing churn distribution
• Bar Chart for customer segmentation
• Bar Chart for age vs churn
• Bar Chart for balance vs risk score
• KPI Cards showing total customers, churn count, and average risk score

6. Key Insights

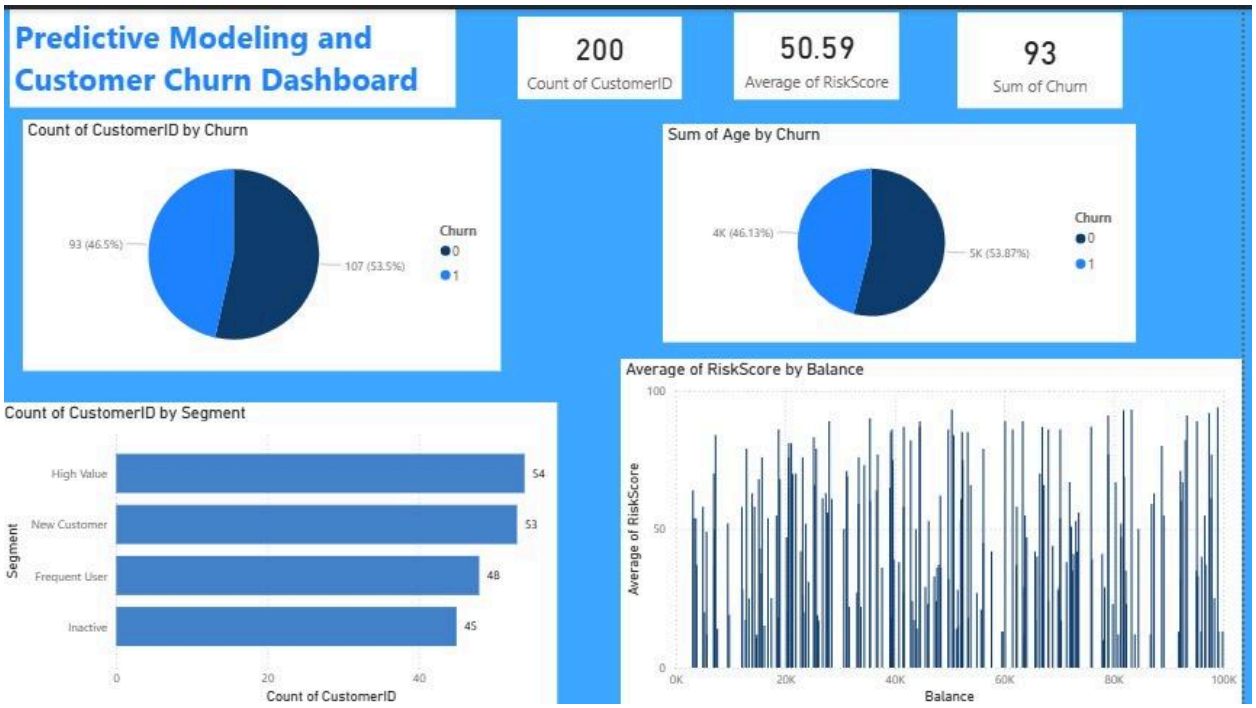
•Inactive customers have the highest churn rate
• Customers with low balance are more likely to churn
• Younger and less active users show higher churn tendency
• High-value customers are more stable

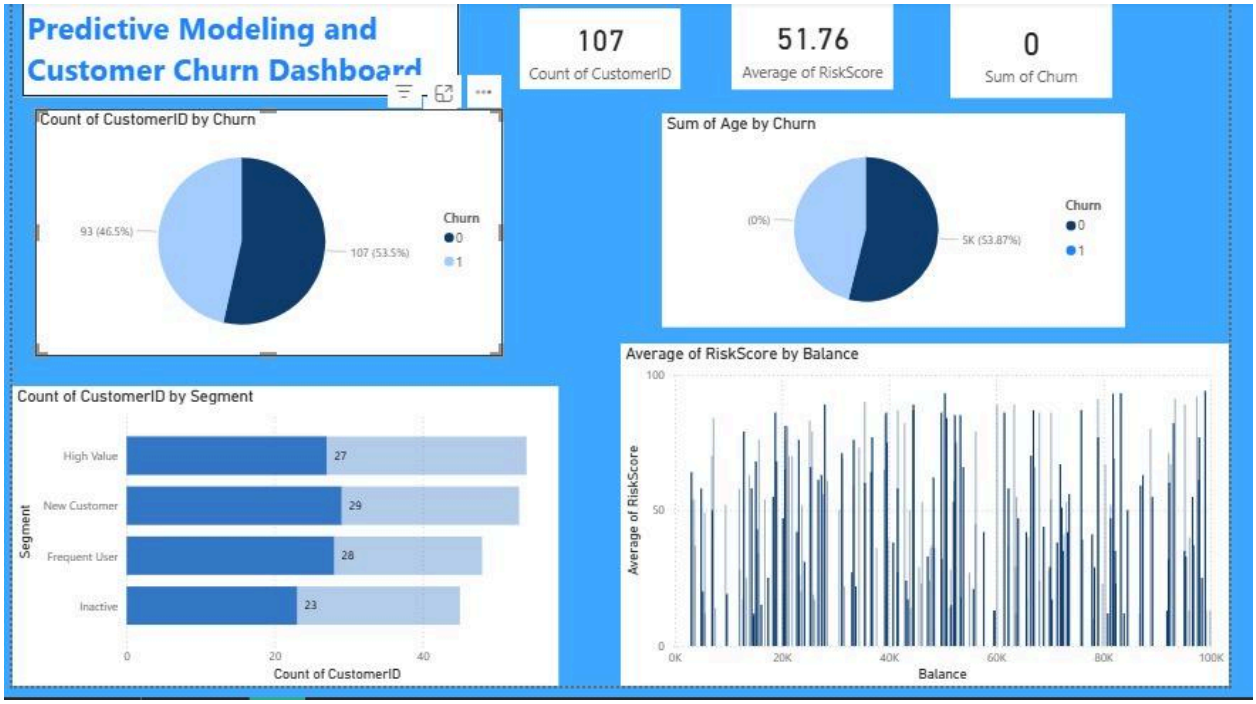
7. Recommendations

•Provide special offers to high-risk customers
• Improve engagement with inactive users
• Offer personalized banking services
Introduce loyalty programs

8. Conclusion

This project helps in identifying customers who are likely to leave the bank. By using customer segmentation and risk scoring, banks can take proactive steps to retain customers and improve their overall performance.




Predictive Modeling and Customer Churn Dashboard
107
Count of CustomerID
51.76
Average of RiskScore
0
Sum of Churn
Count of CustomerID by Churn
93 (46.5%)
107 (53.5%)
Churn
0
1
Sum of Age by Churn
(0%)
5K (53.87%)
Churn
0
1
Count of CustomerID by Segment
High Value
New Customer
Frequent User
Inactive
27
29
28
23
Segment
0
20
40
Count of CustomerID
Average of RiskScore by Balance
100
50
0
Average of RiskScore
0K
20K
40K
60K
80K
100K
Balance

Predictive Modeling and Customer Churn Dashboard
200
Count of CustomerID
50.59
Average of RiskScore
93
Sum of Churn
Count of CustomerID by Churn
93 (46.5%)
107 (53.5%)
Churn
0
1
Sum of Age by Churn
4K (46.13%)
5K (53.87%)
Churn
0
1
Count of CustomerID by Segment
High Value
New Customer
Frequent User
Inactive
54
53
48
45
Segment
0
20
40
Count of CustomerID
Average of RiskScore by Balance
100
50
0
Average of RiskScore
0K
20K
40K
60K
80K
100K
Balance